

Concept Classification Matrix

Stream	Subtheme	Contents (concepts)	Representative papers
Public procurement	Procurement as strategic instrument	Demand-side innovation; mission-oriented procurement; public value; strategic state demand; policy-driven procurement	Edler & Georghiou, 2007; Edquist & Zabala-Iturriagoitia, 2012; Uyarra et al., 2020; Grandia & Meehan, 2017; Patrucco et al., 2024; Kundu et al., 2025; [Damanpour & Schneider, 2009; Begzhan et al., 2026]
	Sustainable and circular procurement	Green public procurement; circular procurement; sustainability operationalization; criteria development; intermediation; environmental performance criteria	Bratt et al., 2013; Rainville, 2021; Ahsan & Rahman, 2017; Grandia & Voncken, 2019; Sparrevik et al., 2018; Lindfors, 2021; Rosell, 2021; Han et al., 2024; Montalban-Domingo et al., 2023; [Tsaousis & Papafloratos, 2026]
	Procurement as formal decision and compliance domain	Specifications; exclusion grounds; qualification requirements; award criteria; contract clauses; auditability; legal defensibility; governance capacity	Patrucco et al., 2024; Uyarra et al., 2020; Stoica et al., 2018; Pariso & Marino, 2025; Yanuarisa et al., 2025; Begzhan et al., 2026; [Ghezzi, 2022; Zitiello et al., 2026]
	Organizational and implementation constraints	Capability deficits; fragmented digitalization; weak institutional support; procedural complexity; adoption barriers; platformization; ICT procurement tensions	Ahsan & Rahman, 2017; Uyarra et al., 2020; Patrucco et al., 2024; Stoica et al., 2018; Ghezzi, 2022; Mavidis et al., 2024; Bunesco & Serban, 2025; [Pellegrini et al., 2021; Zitiello et al., 2026]
DPP / product-data infrastructure	Regulatory and institutional framing	ESPR; delegated acts; product priorities; staged rollout; passport service providers; conformity infrastructure	Council of the European Union, 2023; European Commission, 2024; European Commission, 2023; [Council of the European Union, 2023 ADD 1 REV 1; Knothe et al., 2025]
	Ecosystem perspective	Product-passport ecosystem; stakeholder capabilities; access; trust; coordination; ecosystem requirements; impact pathways	King et al., 2023; Chaudhuri et al., 2025; Kemmner et al., 2025; Heess et al., 2024; Christensen et al., 2025; [Otto & Jarke, 2019; Kunz et al., 2024]
	Data content and lifecycle relevance	Product identity; material composition; repairability; maintenance; lifecycle data; provenance; circularity information; end-of-use strategies	Jensen et al., 2023; Berger et al., 2022; Christensen et al., 2025; Morella et al., 2025; Petrik et al., 2025; Bruno et al., 2026; [Zhao & Liu, 2026; Heess et al., 2024]
	Governance and stewardship	Data governance; access rights; validation; accountability; long-term data sustainability; provider governance; trusted sharing	Abraham et al., 2019; Rolland et al., 2023; European Commission, 2024; Kemmner et al., 2025; Heess et al., 2024; [Fabianek et al., 2025; Otto & Jarke, 2019]
	Semantics and distributed exchange	Semantic interoperability; UNTP semantics; AAS submodels; ontologies; decentralized architectures; connector-based exchange; reference architectures	Havur et al., 2025; Hulea et al., 2025; Fabianek et al., 2025; Kebede et al., 2024; Gianvincenzi et al., 2025; Greiner et al., 2024; Mateo-Casali et al., 2025; Morellini et al., 2025; Kannappan et al., 2025; [Juhlin et al., 2024]

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EA / EAM	Foundational architecture logic	Multi-view architecture; layered design; architecture as coordination logic; operating model; public-sector architecture; interoperability frameworks	Zachman, 1987; Ross et al., 2006; The Open Group, 2022; Boh & Yellin, 2006; Guijarro, 2007; Janssen & Hjort-Madsen, 2007; [European Commission, 2017]
	EA value creation	Alignment; standardization; architectural insight; shared understanding; transformation support; architecture service capability; resource-based value	Tamm et al., 2011; Foorthuis et al., 2016; Schmidt & Buxmann, 2011; Lange et al., 2016; Shanks et al., 2018; van de Wetering, 2020
	Governance and organizational anchoring	Principles; standards; decision rights; organizational embedding; participation; compliance; architecture governance	Boh & Yellin, 2006; Lange et al., 2016; Schmidt & Buxmann, 2011; Shanks et al., 2018; Foorthuis et al., 2016; [The Open Group, 2022]
	Public-sector and sustainability-oriented architecture	Accountability; cross-organizational coordination; ecosystem orientation; legacy heterogeneity; sustainable transformation; public-sector software architecture	European Commission, 2017; Guijarro, 2007; Ghezzi et al., 2024; Laumann & Tambo, 2018; Setala et al., 2021; Alves et al., 2016; Garcia-Escallon et al., 2021; [Dias et al., 2026]
Bridge literature	Enterprise interoperability	Legal interoperability; organizational interoperability; semantic interoperability; technical interoperability; enterprise integration	Chen et al., 2008; European Commission, 2017; Guijarro, 2007; Knothe et al., 2025; [Juhlin et al., 2024]
	Platforms and infrastructures	Digital platforms; sociotechnical coordination; multi-actor interaction; infrastructure logic; ecosystem participation	de Reuver et al., 2018; Constantinides et al., 2018; Otto & Jarke, 2019; Kolehmainen et al., 2025; [Pariso & Marino, 2025]
	Dataspaces and federated exchange	Distributed data access; data sovereignty; federated governance; mediated participation; dataspace connectors; marketplaces	Franklin et al., 2005; Otto & Jarke, 2019; Hulea et al., 2025; Kunz et al., 2024; [Kemmner et al., 2025; Greiner et al., 2024]
	DPP-specific bridging mechanisms	Connector-based exchange; mediated access; compliance-by-design; provider-based validation; domain-specific modeling; circular transition support	Hulea et al., 2025; Fabianek et al., 2025; European Commission, 2024; Shirvani et al., 2018; Laumann & Tambo, 2018; [Anthony et al., 2018; Funke et al., 2025]